

# From Sound to Shape: Reconstructing Vocal Dynamics with Time-Delay Embedding

## (Dynamical structures in bird songs)

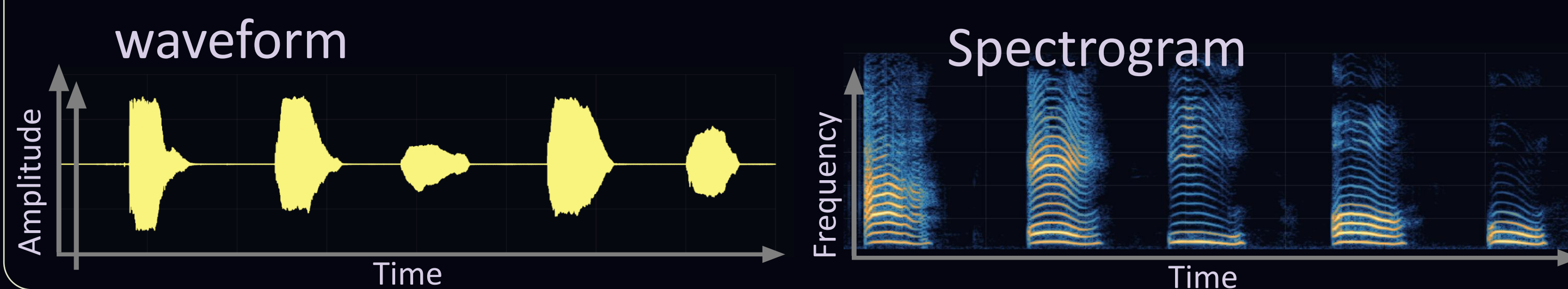


**Navyanth Kusampudi**  
PostDoc, Max-Planck-Institut for Sustainable Materials GmbH, Germany

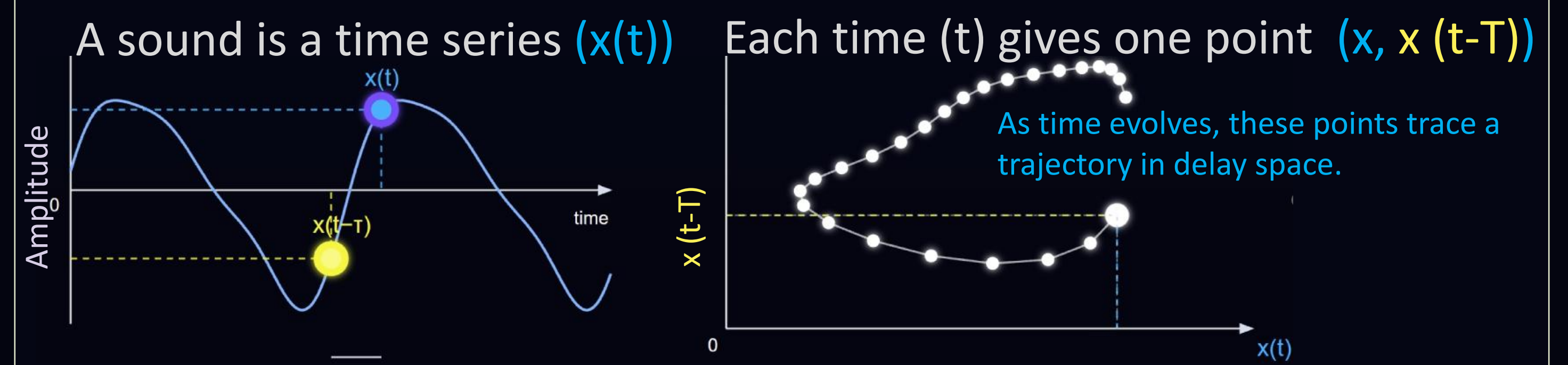


### 1 Why sound as dynamics?

- Waveforms and spectrograms describe the acoustic signal.
- But vocal production is a dynamical process.
- Can sound reveal structure in the dynamics that generated it?



### 2 From sound to shape



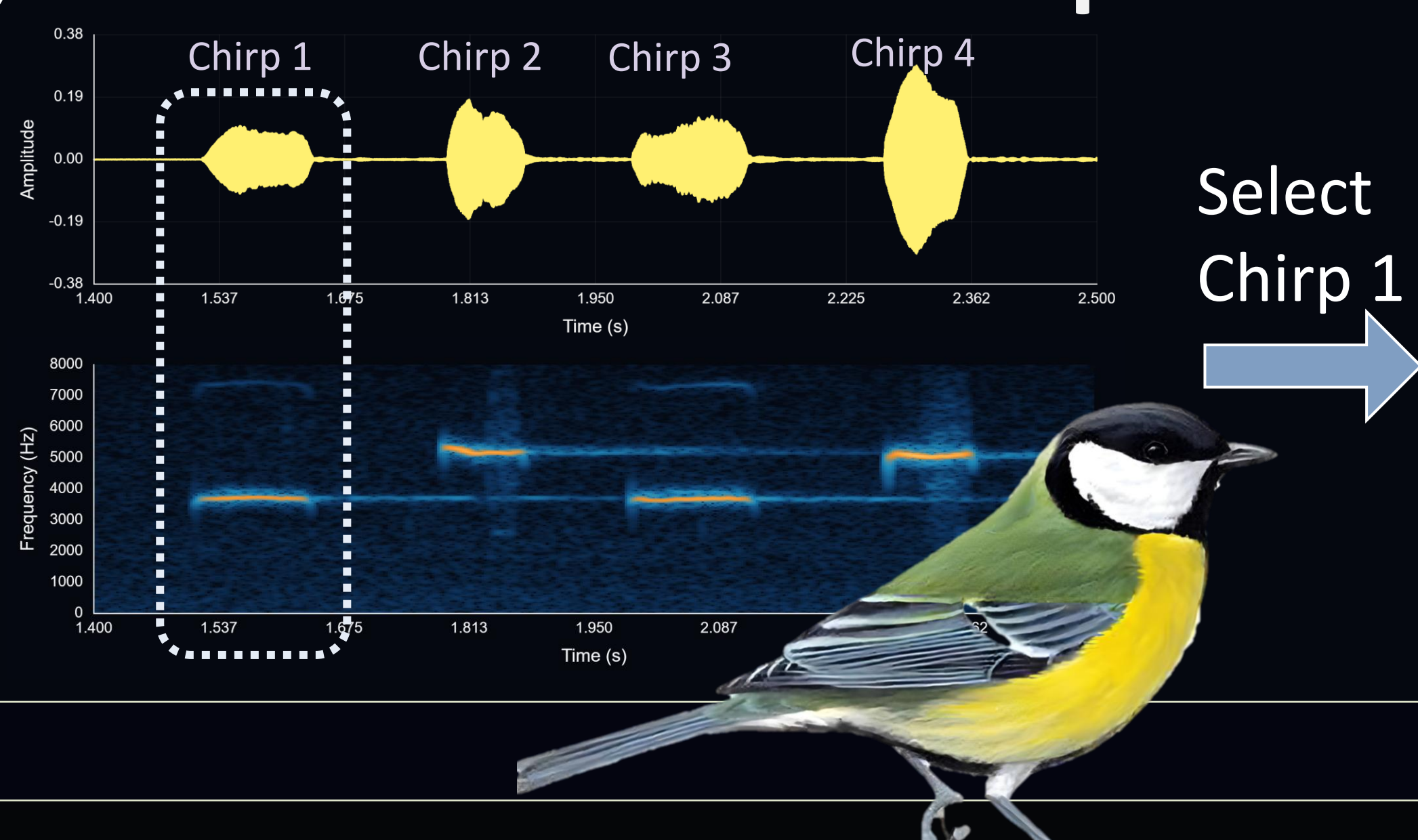
Time-delay embedding transforms a one-dimensional sound signal into a geometric representation of its temporal dynamics.

### 3 What do these shapes mean

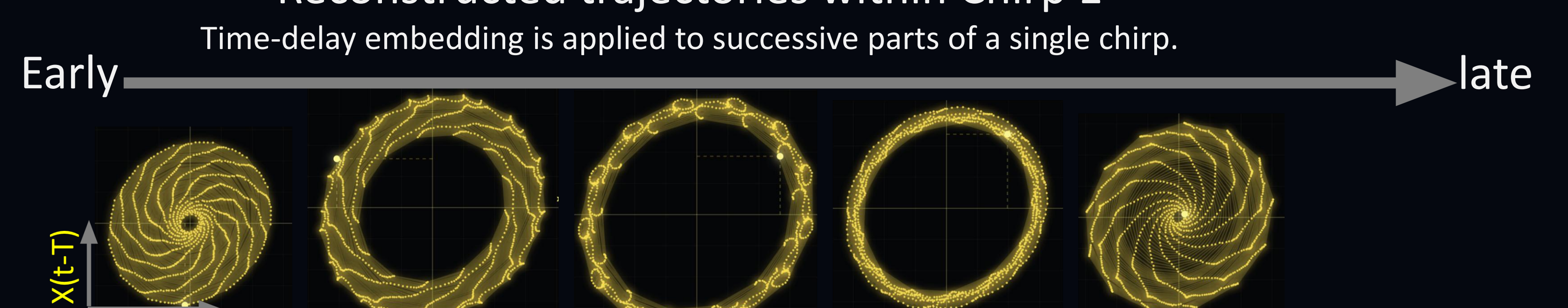


- Different temporal structure produces a different reconstructed shape
- These shapes can reveal recurrence, transitions, and multiple timescales.

### 4 Bird vocalization example

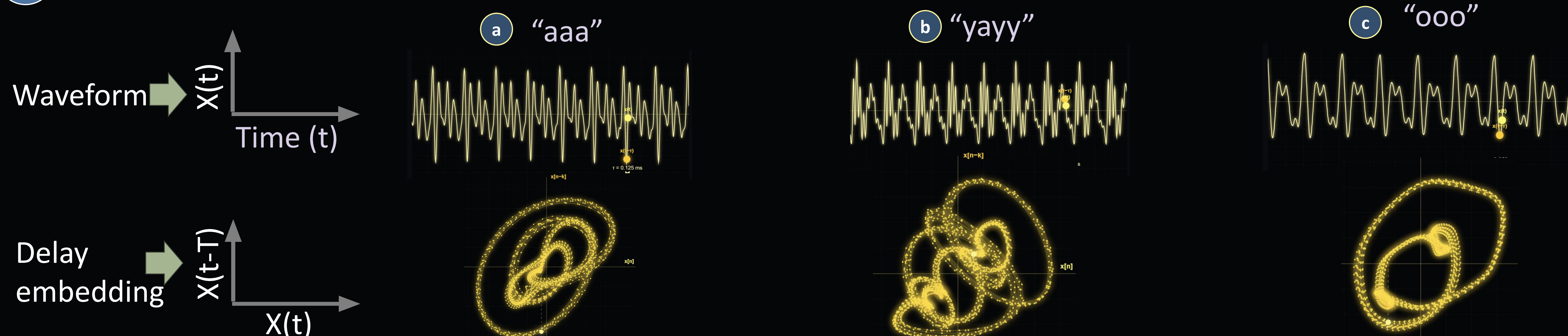


#### Reconstructed trajectories within Chirp 1



- The trajectory shows changing vocal dynamics even within one chirp
- Could reconstructed vocal dynamics provide a dynamical fingerprint of a call, individual, or species?

### 5 Adult human vocal sounds



### 6 Communication as inference

Bird A produces a call

Bird B listens and predicts what may come next.

- Communication can emerge through reciprocal prediction and inference over sensory signals. (Friston 2015)
- Hear → Predict → Respond → Coordinate
- Can reconstructed vocal trajectories reveal coordination between communicating birds?

### 7 Take home message

Sound is not only a spectrum – it is a trajectory through time

- Time delay embedding turns sound into geometry
- Reconstructed trajectories provide a complementary way to study vocal dynamics, temporal structure, variation and communication



Takens, F. (1981). Detecting strange attractors in turbulence. *Lecture Notes in Mathematics*, **898**, 366–381.  
Laje, R. & Mindlin, G. B. (2002). Diversity within a birdsong. *Physical Review Letters*, **89**, 288102.  
Friston, K. & Frith, C. (2015). A duet for one. *Consciousness and Cognition*, **36**, 390–405.

